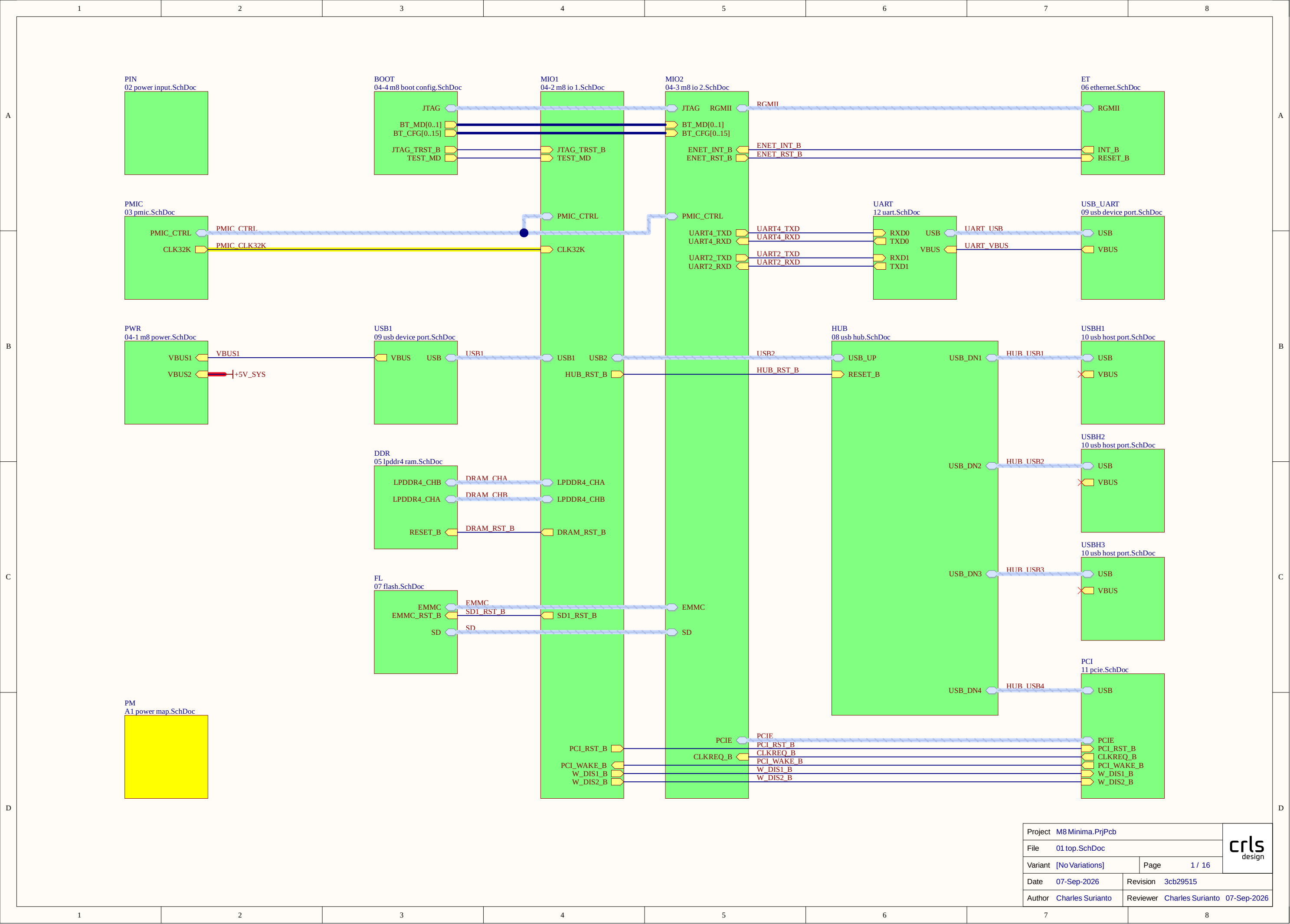
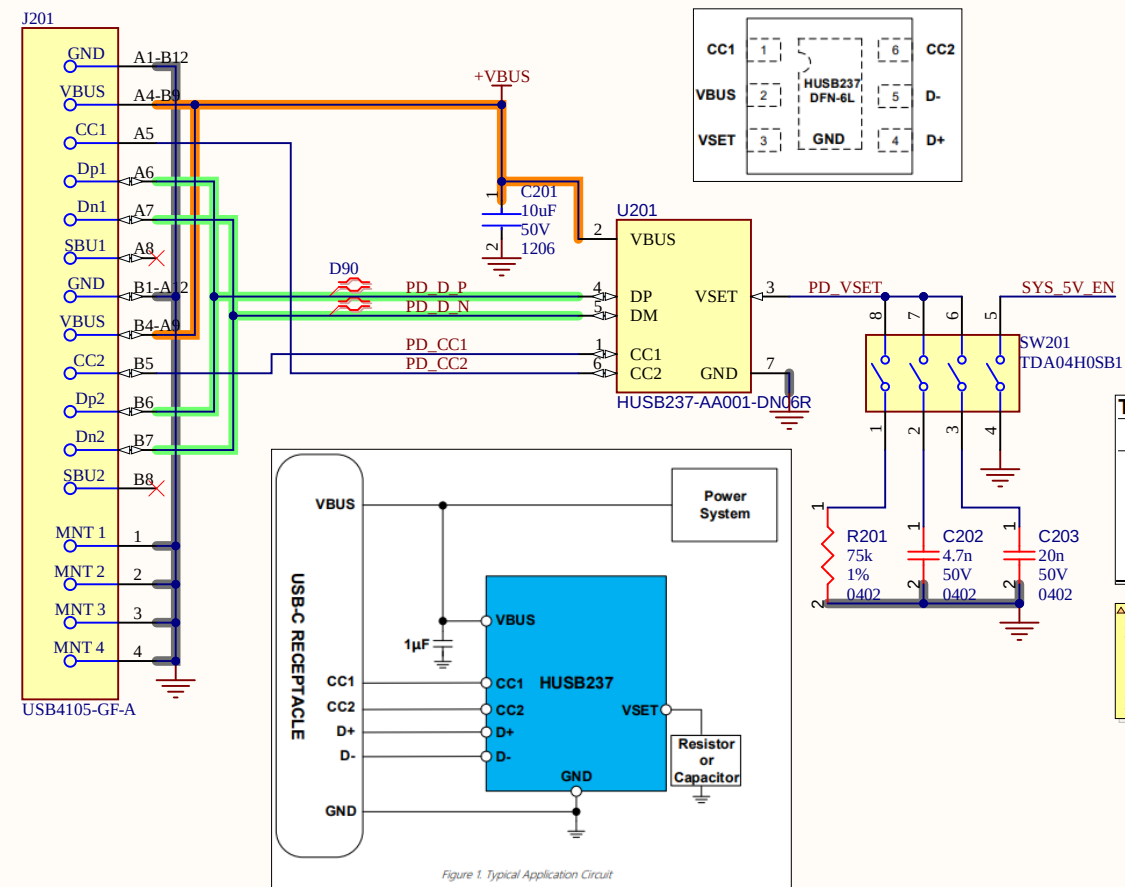


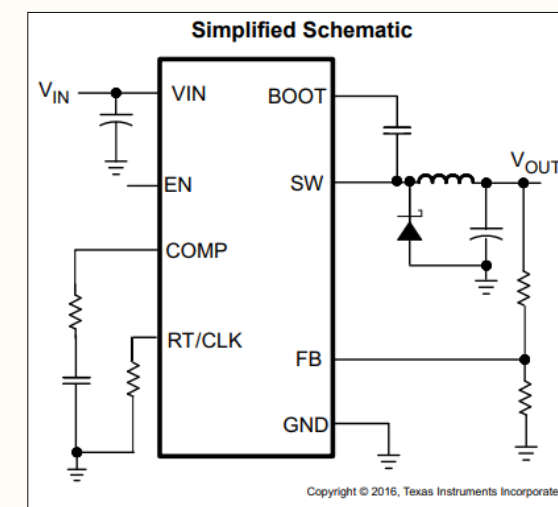
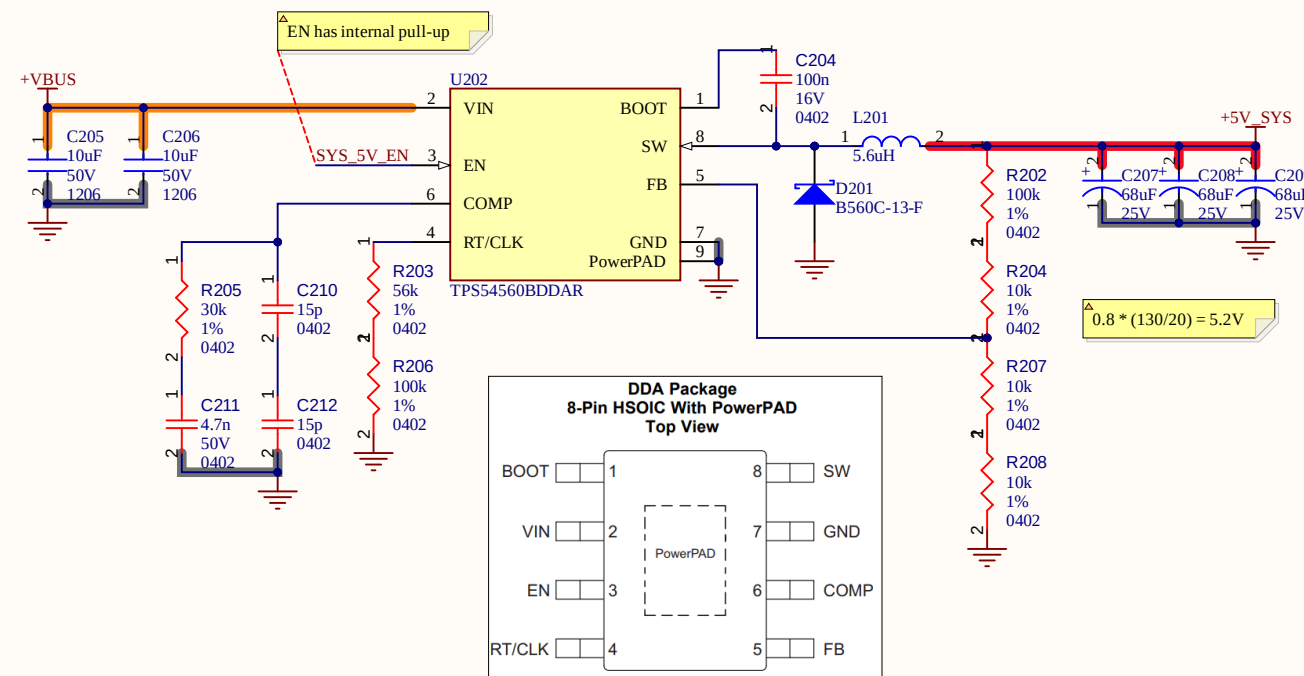


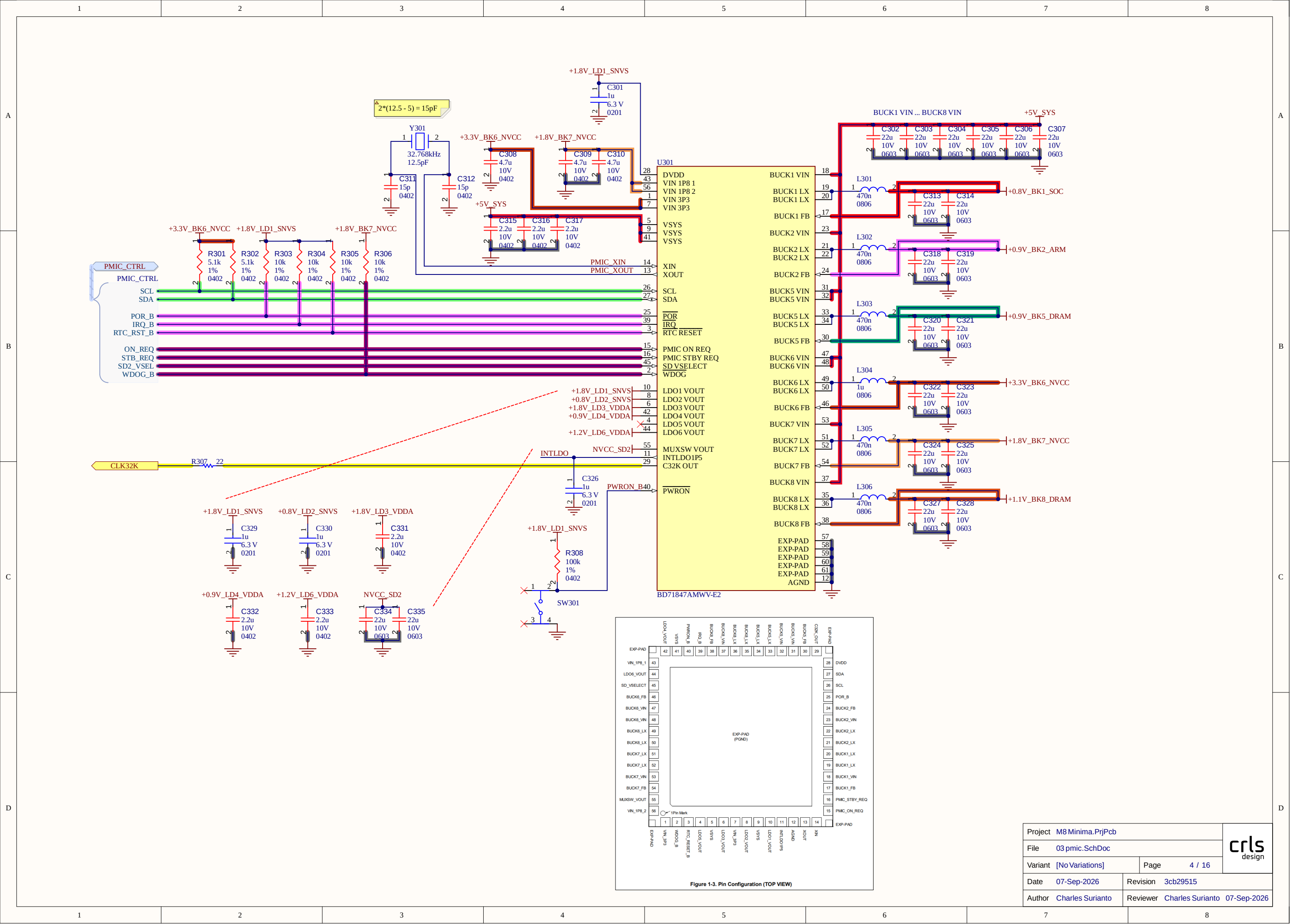


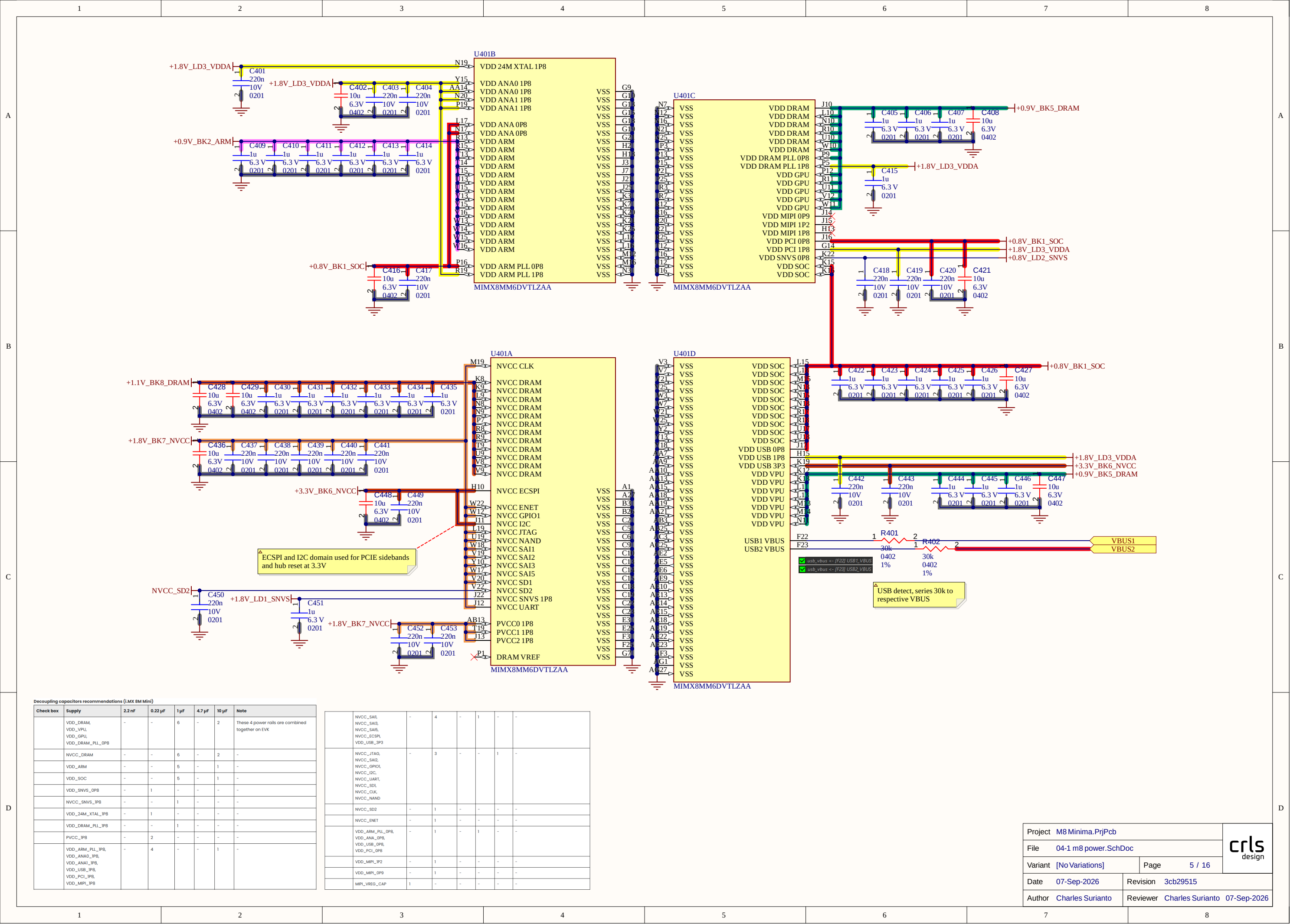
VSET Pin Configuration	Preferred Voltage (V)
0 Ω	5
75 k Ω	9
open	12
4.7 nF	15
20 nF	20

△ DS: Connect a resistor or capacitor with 1% tolerance between VSET and GND to indicate the target RDO voltage value.

*10% cap should suffice since the difference is a factor of 5, provided DC bias is accounted for.







Ball name	LPDDR4
DRAM_DQ0_P	DQ0_1_A
DRAM_DQ0_N	DQ0_1_B
DRAM_DM0	DM0_A
DRAM_DQ00	DQ0_A
DRAM_DQ01	DQ1_A
DRAM_DQ02	DQ2_A
DRAM_DQ03	DQ3_A
DRAM_DQ04	DQ4_A
DRAM_DQ05	DQ5_A

DRAM_DQ06	DQ6_A
DRAM_DQ07	DQ7_A
DRAM_DQ08_P	DQ08_1_A
DRAM_DQ08_N	DQ08_1_B
DRAM_DM1	DM1_A
DRAM_DQ08	DQ08_A
DRAM_DQ09	DQ09_A
DRAM_DQ10	DQ10_A
DRAM_DQ11	DQ11_A
DRAM_DQ12	DQ12_A
DRAM_DQ13	DQ13_A
DRAM_DQ14	DQ14_A
DRAM_DQ15	DQ15_A
DRAM_DQ02_P	DQ02_1_B
DRAM_DQ02_N	DQ02_1_A
DRAM_DM2	DM2_B
DRAM_DQ16	DQ16_B
DRAM_DQ17	DQ17_B
DRAM_DQ18	DQ18_B
DRAM_DQ19	DQ19_B
DRAM_DQ20	DQ20_B
DRAM_DQ21	DQ21_B
DRAM_DQ22	DQ22_B
DRAM_DQ23	DQ23_B
DRAM_DQ03_P	DQ03_1_B
DRAM_DQ03_N	DQ03_1_A
DRAM_DM3	DM3_B
DRAM_DQ24	DQ24_B
DRAM_DQ25	DQ25_B
DRAM_DQ26	DQ26_B
DRAM_DQ27	DQ27_B
DRAM_DQ28	DQ28_B
DRAM_DQ29	DQ29_B

DRAM_DQ30	DQ30_B
DRAM_DQ31	DQ31_B
DRAM_RESET_N	RESET_N
DRAM_ALERT_N	MTEST1
DRAM_AC00	CKE0_A
DRAM_AC01	CKE1_A
DRAM_AC02	CS0_A
DRAM_AC03	CS1_A
DRAM_AC04	CK_1_A
DRAM_AC05	CK_c_A
DRAM_AC06	---
DRAM_AC07	---
DRAM_AC08	CA0_A
DRAM_AC09	CA1_A
DRAM_AC10	CA2_A
DRAM_AC11	CA3_A
DRAM_AC12	CA4_A
DRAM_AC13	CA5_A
DRAM_AC14	---
DRAM_AC15	---
DRAM_AC16	---
DRAM_AC17	---
DRAM_AC19	MTEST
DRAM_AC20	CKE0_B
DRAM_AC21	CKE1_B
DRAM_AC22	CS1_B
DRAM_AC23	CS0_B
DRAM_AC24	CK_1_B
DRAM_AC25	CK_c_B
DRAM_AC26	---
DRAM_AC27	---
DRAM_AC28	CA0_B
DRAM_AC29	CA1_B
DRAM_AC30	CA2_B

DRAM_DQ30	DQ30_B
DRAM_DQ31	DQ31_B
DRAM_RESET_N	RESET_N
DRAM_ALERT_N	MTEST1
DRAM_AC00	CKE0_A
DRAM_AC01	CKE1_A
DRAM_AC02	CS0_A
DRAM_AC03	CS1_A
DRAM_AC04	CK_1_A
DRAM_AC05	CK_c_A
DRAM_AC06	---
DRAM_AC07	---
DRAM_AC08	CA0_A
DRAM_AC09	CA1_A
DRAM_AC10	CA2_A
DRAM_AC11	CA3_A
DRAM_AC12	CA4_A
DRAM_AC13	CA5_A
DRAM_AC14	---
DRAM_AC15	---
DRAM_AC16	---
DRAM_AC17	---
DRAM_AC19	MTEST
DRAM_AC20	CKE0_B
DRAM_AC21	CKE1_B
DRAM_AC22	CS1_B
DRAM_AC23	CS0_B
DRAM_AC24	CK_1_B
DRAM_AC25	CK_c_B
DRAM_AC26	---
DRAM_AC27	---
DRAM_AC28	CA0_B
DRAM_AC29	CA1_B
DRAM_AC30	CA2_B

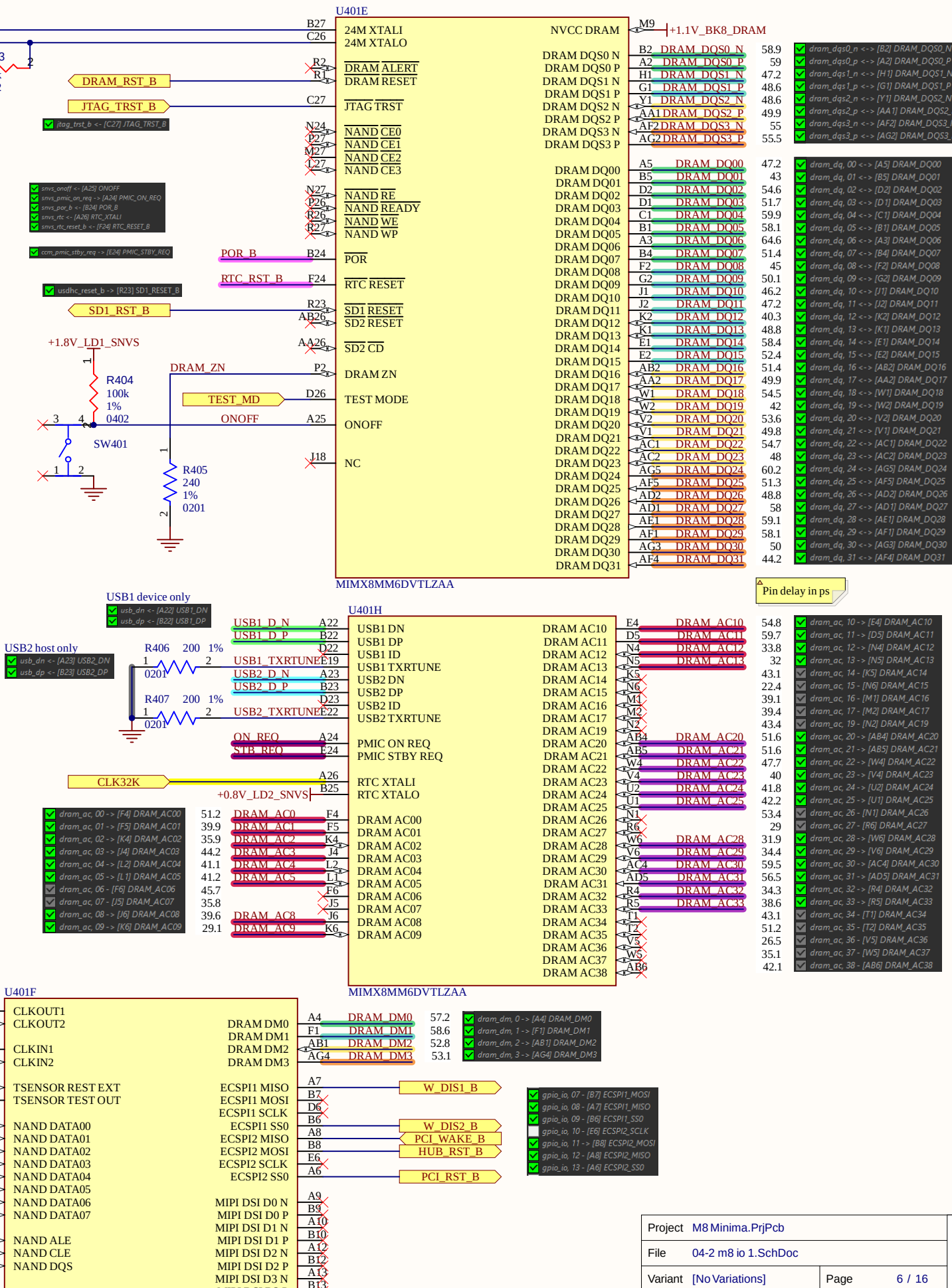
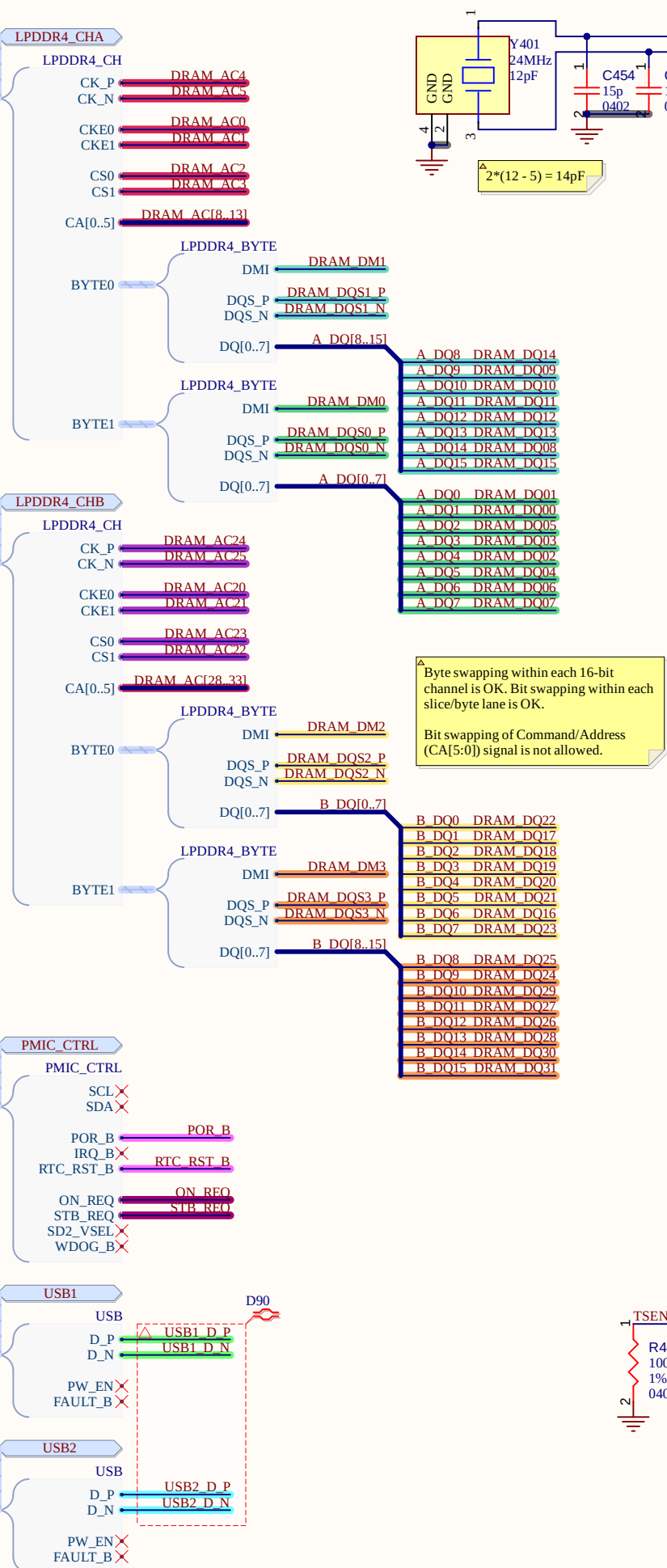
DRAM_DQ30	DQ30_B
DRAM_DQ31	DQ31_B
DRAM_RESET_N	RESET_N
DRAM_ALERT_N	MTEST1
DRAM_AC00	CKE0_A
DRAM_AC01	CKE1_A
DRAM_AC02	CS0_A
DRAM_AC03	CS1_A
DRAM_AC04	CK_1_A
DRAM_AC05	CK_c_A
DRAM_AC06	---
DRAM_AC07	---
DRAM_AC08	CA0_A
DRAM_AC09	CA1_A
DRAM_AC10	CA2_A
DRAM_AC11	CA3_A
DRAM_AC12	CA4_A
DRAM_AC13	CA5_A
DRAM_AC14	---
DRAM_AC15	---
DRAM_AC16	---
DRAM_AC17	---
DRAM_AC19	MTEST
DRAM_AC20	CKE0_B
DRAM_AC21	CKE1_B
DRAM_AC22	CS1_B
DRAM_AC23	CS0_B
DRAM_AC24	CK_1_B
DRAM_AC25	CK_c_B
DRAM_AC26	---
DRAM_AC27	---
DRAM_AC28	CA0_B
DRAM_AC29	CA1_B
DRAM_AC30	CA2_B

DRAM_DQ30	DQ30_B
DRAM_DQ31	DQ31_B
DRAM_RESET_N	RESET_N
DRAM_ALERT_N	MTEST1
DRAM_AC00	CKE0_A
DRAM_AC01	CKE1_A
DRAM_AC02	CS0_A
DRAM_AC03	CS1_A
DRAM_AC04	CK_1_A
DRAM_AC05	CK_c_A
DRAM_AC06	---
DRAM_AC07	---
DRAM_AC08	CA0_A
DRAM_AC09	CA1_A
DRAM_AC10	CA2_A
DRAM_AC11	CA3_A
DRAM_AC12	CA4_A
DRAM_AC13	CA5_A
DRAM_AC14	---
DRAM_AC15	---
DRAM_AC16	---
DRAM_AC17	---
DRAM_AC19	MTEST
DRAM_AC20	CKE0_B
DRAM_AC21	CKE1_B
DRAM_AC22	CS1_B
DRAM_AC23	CS0_B
DRAM_AC24	CK_1_B
DRAM_AC25	CK_c_B
DRAM_AC26	---
DRAM_AC27	---
DRAM_AC28	CA0_B
DRAM_AC29	CA1_B
DRAM_AC30	CA2_B

DRAM_DQ30	DQ30_B
DRAM_DQ31	DQ31_B
DRAM_RESET_N	RESET_N
DRAM_ALERT_N	MTEST1
DRAM_AC00	CKE0_A
DRAM_AC01	CKE1_A
DRAM_AC02	CS0_A
DRAM_AC03	CS1_A
DRAM_AC04	CK_1_A
DRAM_AC05	CK_c_A
DRAM_AC06	---
DRAM_AC07	---
DRAM_AC08	CA0_A
DRAM_AC09	CA1_A
DRAM_AC10	CA2_A
DRAM_AC11	CA3_A
DRAM_AC12	CA4_A
DRAM_AC13	CA5_A
DRAM_AC14	---
DRAM_AC15	---
DRAM_AC16	---
DRAM_AC17	---
DRAM_AC19	MTEST
DRAM_AC20	CKE0_B
DRAM_AC21	CKE1_B
DRAM_AC22	CS1_B
DRAM_AC23	CS0_B
DRAM_AC24	CK_1_B
DRAM_AC25	CK_c_B
DRAM_AC26	---
DRAM_AC27	---
DRAM_AC28	CA0_B
DRAM_AC29	CA1_B
DRAM_AC30	CA2_B

DRAM_DQ30	DQ30_B
DRAM_DQ31	DQ31_B
DRAM_RESET_N	RESET_N
DRAM_ALERT_N	MTEST1
DRAM_AC00	CKE0_A
DRAM_AC01	CKE1_A
DRAM_AC02	CS0_A
DRAM_AC03	CS1_A
DRAM_AC04	CK_1_A
DRAM_AC05	CK_c_A
DRAM_AC06	---
DRAM_AC07	---
DRAM_AC08	CA0_A
DRAM_AC09	CA1_A
DRAM_AC10	CA2_A
DRAM_AC11	CA3_A
DRAM_AC12	CA4_A
DRAM_AC13	CA5_A
DRAM_AC14	---
DRAM_AC15	---
DRAM_AC16	---
DRAM_AC17	---
DRAM_AC19	MTEST
DRAM_AC20	CKE0_B
DRAM_AC21	CKE1_B
DRAM_AC22	CS1_B
DRAM_AC23	CS0_B
DRAM_AC24	CK_1_B
DRAM_AC25	CK_c_B
DRAM_AC26	---
DRAM_AC27	---
DRAM_AC28	CA0_B
DRAM_AC29	CA1_B
DRAM_AC30	CA2_B

DRAM_DQ30	DQ30_B
DRAM_DQ31	DQ31_B
DRAM_RESET_N	RESET_N
DRAM_ALERT_N	MTEST1
DRAM_AC00	CKE0_A
DRAM_AC01	CKE1_A
DRAM_AC02	CS0_A
DRAM_AC03	CS1_A
DRAM_AC04	CK_1_A
DRAM_AC05	CK_c_A
DRAM_AC06	---
DRAM_AC07	---
DRAM_AC08	CA0_A
DRAM_AC09	CA1_A
DRAM_AC10	CA2_A
DRAM_AC11	CA3_A
DRAM_AC12	CA4_A
DRAM_AC13	CA5_A
DRAM_AC14	---
DRAM_AC15	---
DRAM_AC16	---
DRAM_AC17	---
DRAM_AC19	MTEST
DRAM_AC20	CKE0_B
DRAM_AC21	CKE1_B
DRAM_AC22	CS1_B
DRAM_AC23	CS0_B
DRAM_AC24	CK_1_B
DRAM_AC25	CK_c_B
DRAM_AC26	---
DRAM_AC27	---
DRAM_AC28	CA0_B
DRAM_AC29	CA1_B
DRAM_AC30	CA2_B



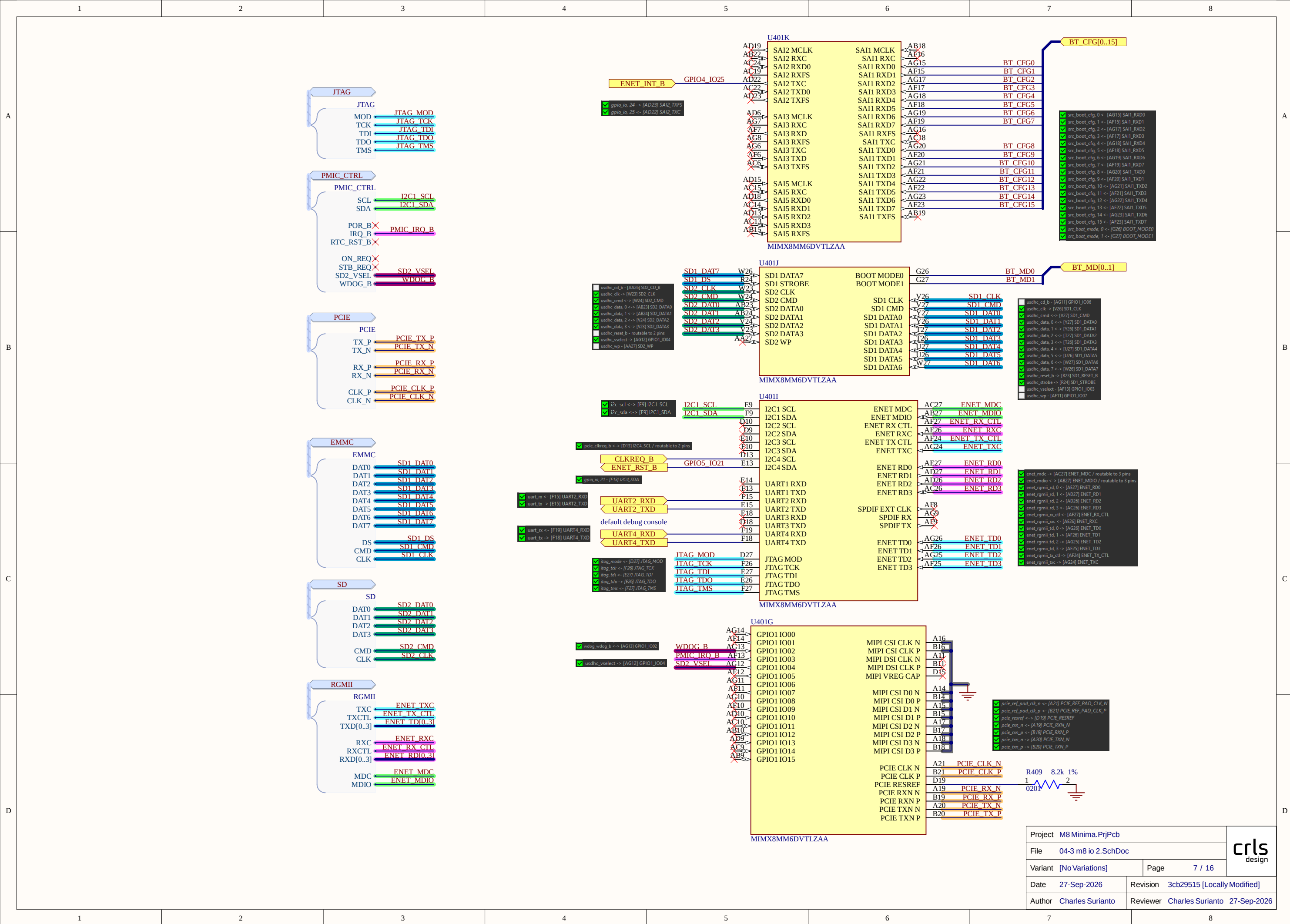


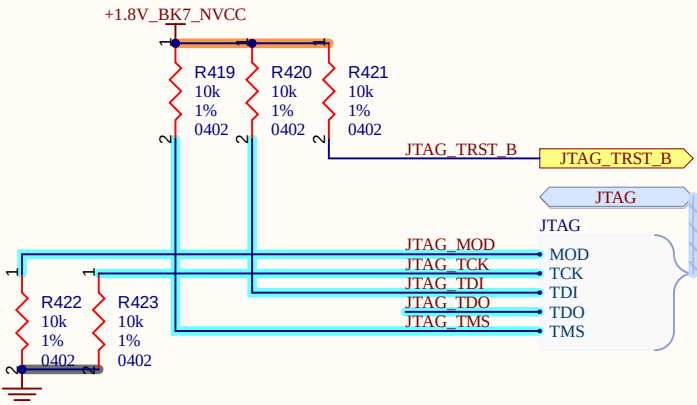
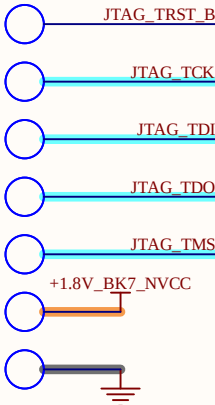
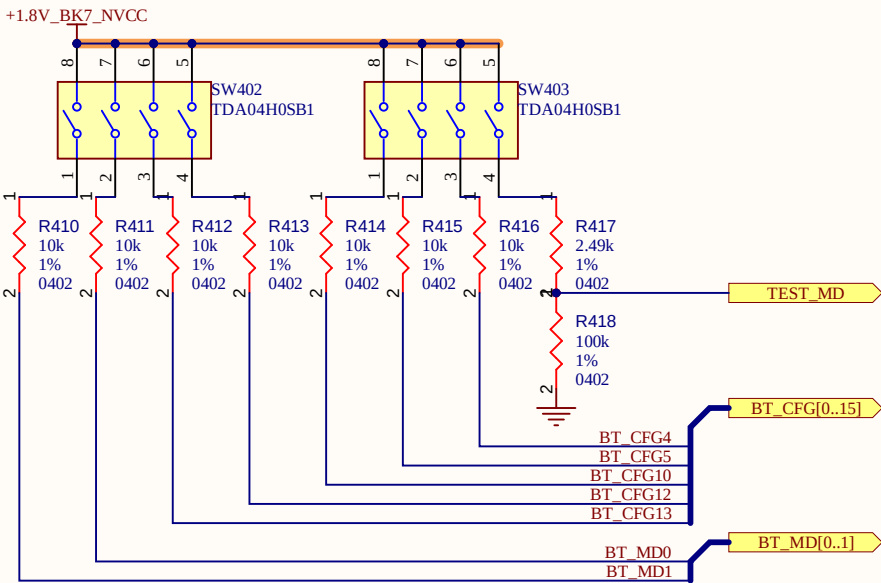
Table 6-1. Boot MODE pin settings	
BOOT_MODE[1:0]	Boot Type
00	Boot From Fuses
01	Serial Downloader
10	Internal Boot
11	Reserved

Table 6-9. Boot device selection	
BOOT_CFG[14:12]	Boot device
001	SD/eSD
010	MMC/eMMC
011	NAND
100	Serial NOR boot via FlexSPI
110	Serial (SPI) NOR

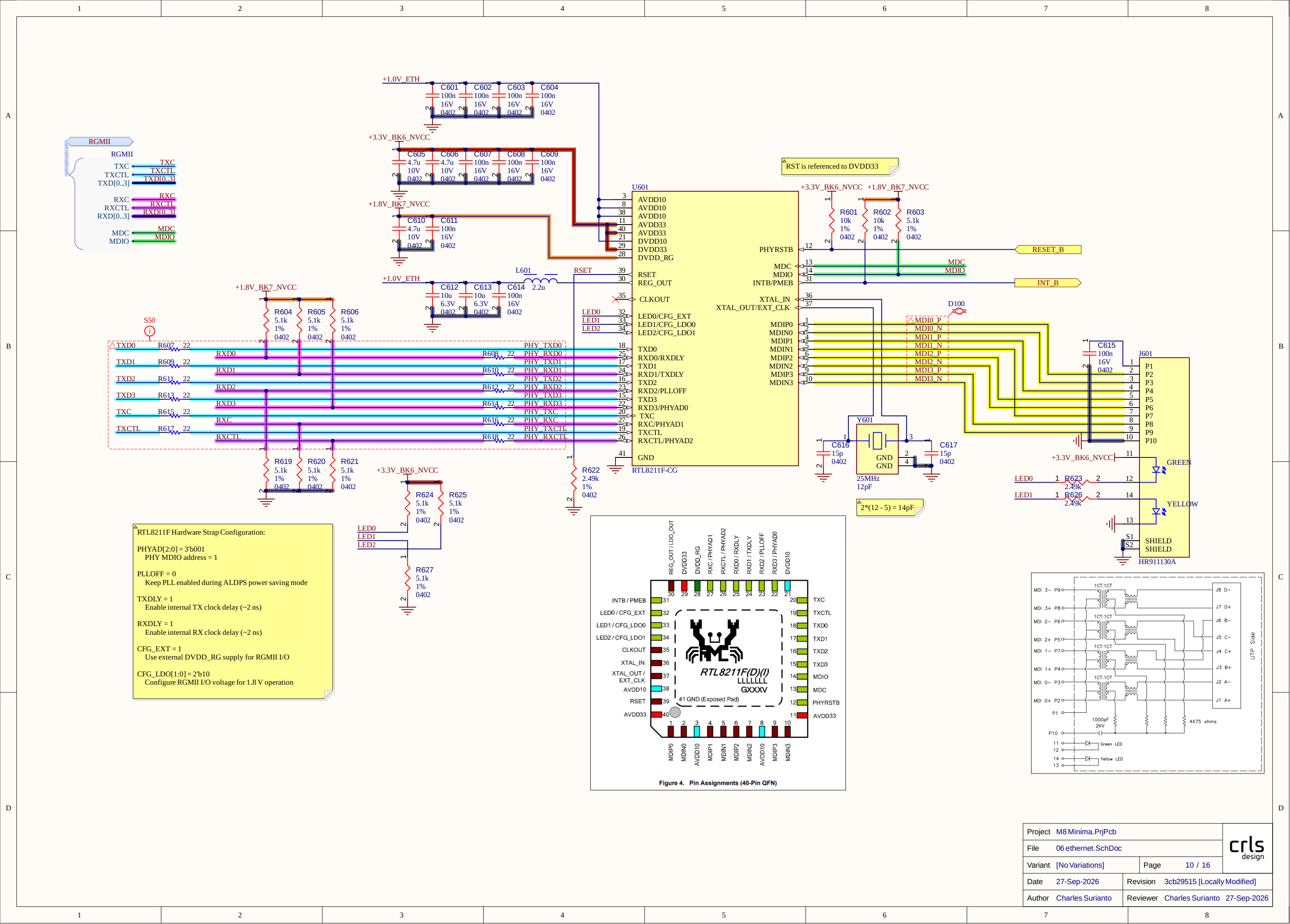
Table 6-42. SD/eSD Boot Fusemap								
Addr	7	6	5	4	3	2	1	0
0x470[15:8]	Reserved	Boot Device Select			Port Select: 00 - SD1 01 - SD2 10 - SD3		Power Cycle Enable 0 - Disable 1 - Enable	SD Loopback Clock Source SEL (SDR50 and SDR104 Only) 0 - Through SD pad 1 - Direct
0x470[7:0]	Fast Boot: 0 - Regular 1 - Fast Boot	Reserved		Bus Width: 0 - 1-bit 1 - 4-bit	Speed 000 - Normal/SDR12 001 - High/SDR25 010 - SDR50 011 - SDR104 101 - Reserved for DDR50 Others - Reserved			Reserved

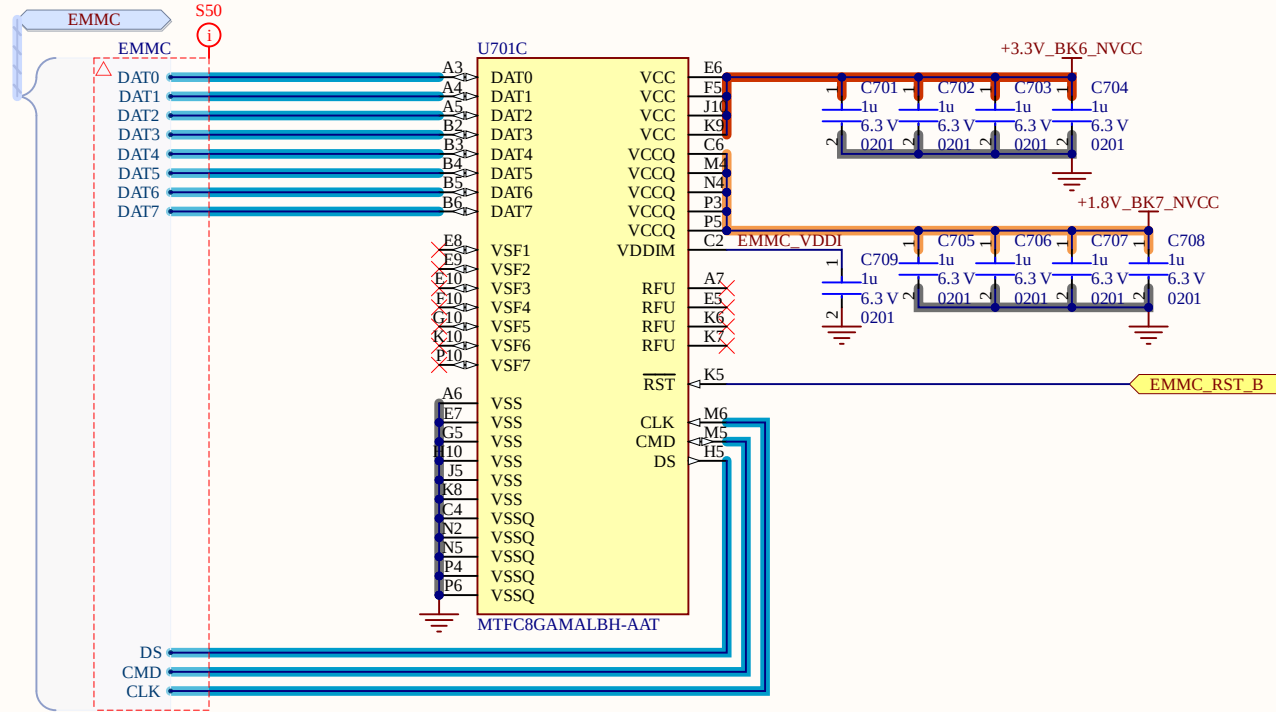
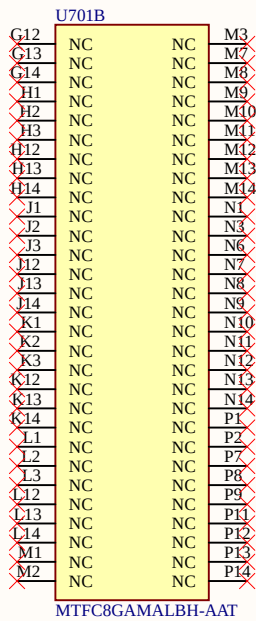
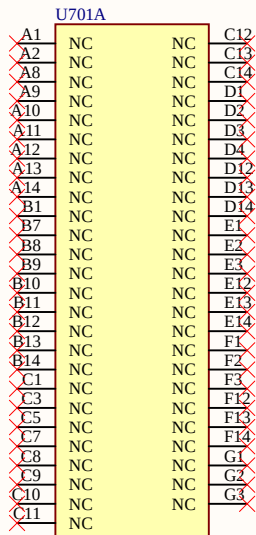
Table 6-43. MMC/eMMC Boot Fusemap								
Addr	7	6	5	4	3	2	1	0
0x470[15:8]	Reserved	Boot Device Select			Port Select: 00 - SD1 01 - SD2 10 - SD3		Power Cycle Enable 0 - Disable 1 - Enable	SD Loopback Clock Source SEL (SDR50 and SDR104 Only) 0 - Through SD pad 1 - Direct
0x470[7:0]	Fast Boot: 0 - Regular 1 - Fast Boot	Bus Width: 000 - 1-bit 001 - 4-bit 010 - 8-bit 101 - 4-bit DDR (MMC 4.4) 110 - 8-bit DDR (MMC 4.4) Else - Reserved			Speed 00 - Normal 01 - High 10 - Reserved for HS200 11 - Reserved		USDHC IO VOLTAGE SELECTION N For Normal Boot Mode 0 - 3.3V 1 - 1.8V	USDHC IO VOLTAGE SELECTION N For Manufactur e Mode 0 - 3.3V 1 - 1.8V

11|XXXXX|1 => JTAG
01|XXXXX|0 => USB1 OTG
10|10010|0 => SD1 eMMC 8bit
10|011X1|0 => SD2 microSD 4bit



Recommended board terminations for Boundary-scan			
Signal	I/O type	External termination	Comments
BOOT_MODE0	Input	Pull-up	BOOT_MODE0, BOOT_MODE1, JTAG_MOD, and TEST_MODE must be at 1101 to enter in Boundary-scan mode.
BOOT_MODE1	Input	Pull-up	
JTAG_MOD	Input	Pull-down	
TEST_MODE	Input	Pull-up	
GPIO1_IO02 (WD0G_B)	Input	Pull-up	External pull up resistor (100 Kilohms) must support boundary-scan mode.





[Table 2] 153 Ball Information

Pin NO	Name
A3	DAT0
A4	DAT1
A5	DAT2
B2	DAT3
B3	DAT4
B4	DAT5
B5	DAT6
B6	DAT7
K5	RSTN
C6	VDD
M4	VDD
P3	VDD
P5	VDD
E6	VDDF
F5	VDDF
J10	VDDF
K9	VDDF
C2	VDDI
M5	CMD
H5	Data Strobe
M6	CLK
J5	VSS
C4	VSS
E7	VSS
G5	VSS
H10	VSS
K8	VSS
N2	VSS
N5	VSS
P4	VSS
P6	VSS

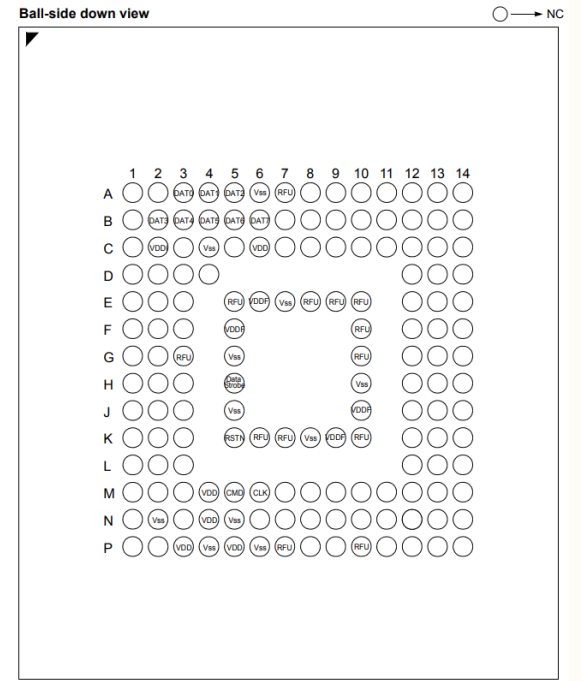
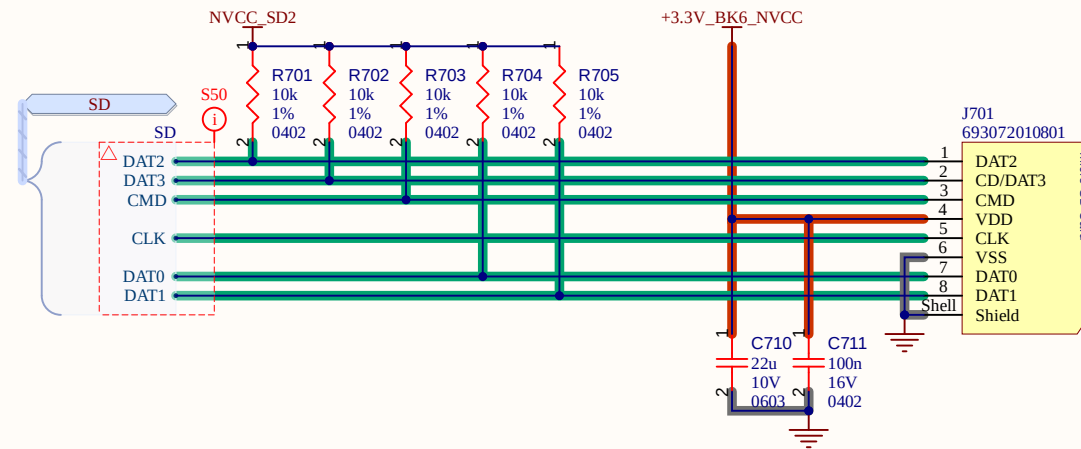
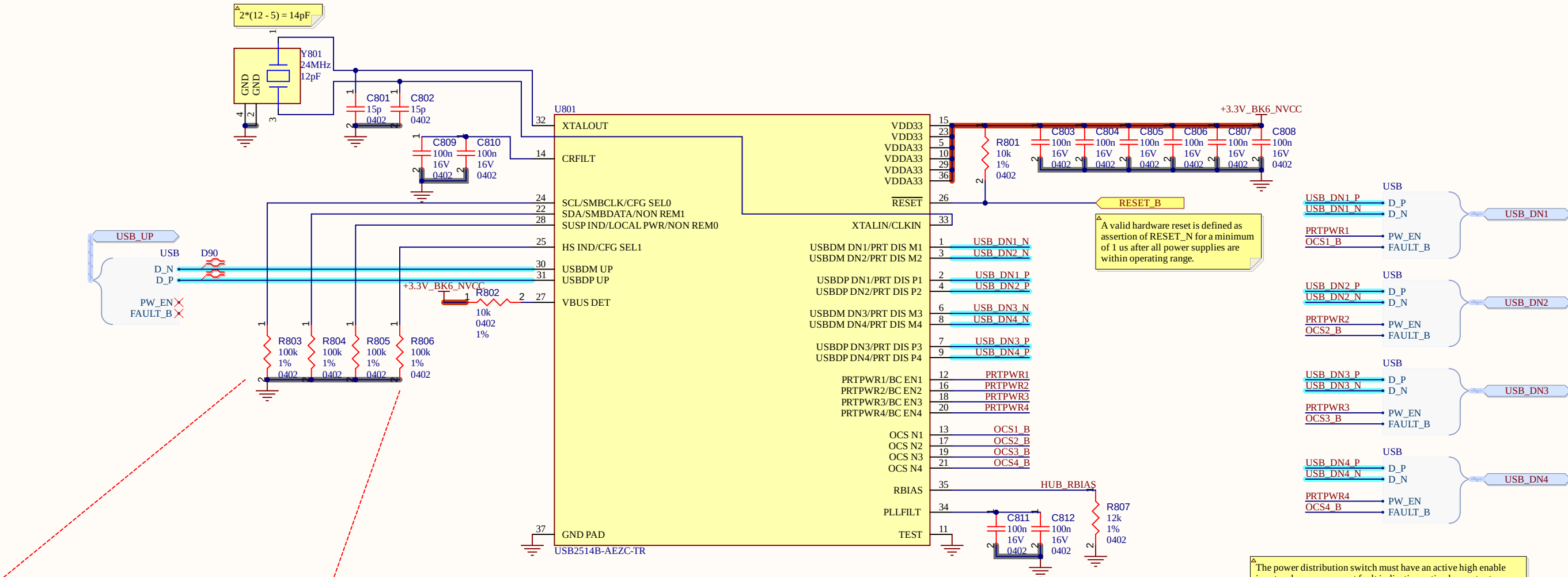


Figure 1. 153-FBGA

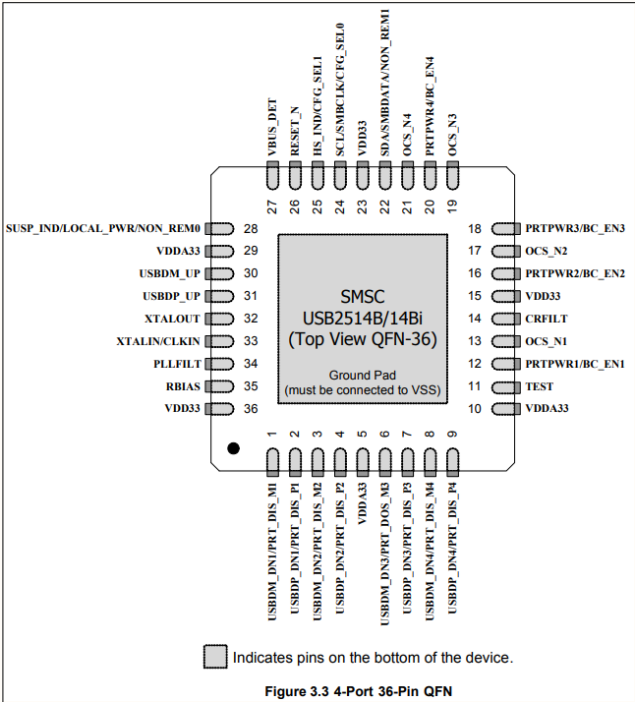


PIN No	NAME	DESCRIPTION
1	DAT2	Connector data line 2
2	CD/DAT3	CARD DETECTION
3	CMD	COMMAND/ANSWER
4	Vdd	SUPPLY VOLTAGE
5	CLK	CLOCK
6	Vss	GND
7	DAT0	Connector data line 0
8	DAT1	Connector data line 1

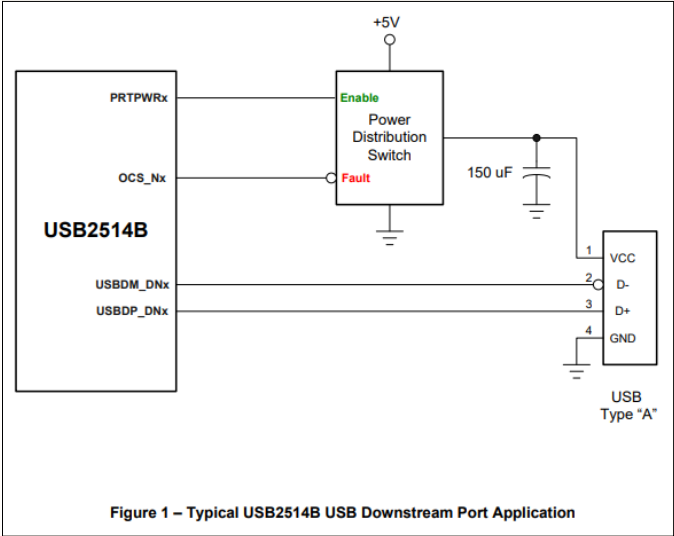


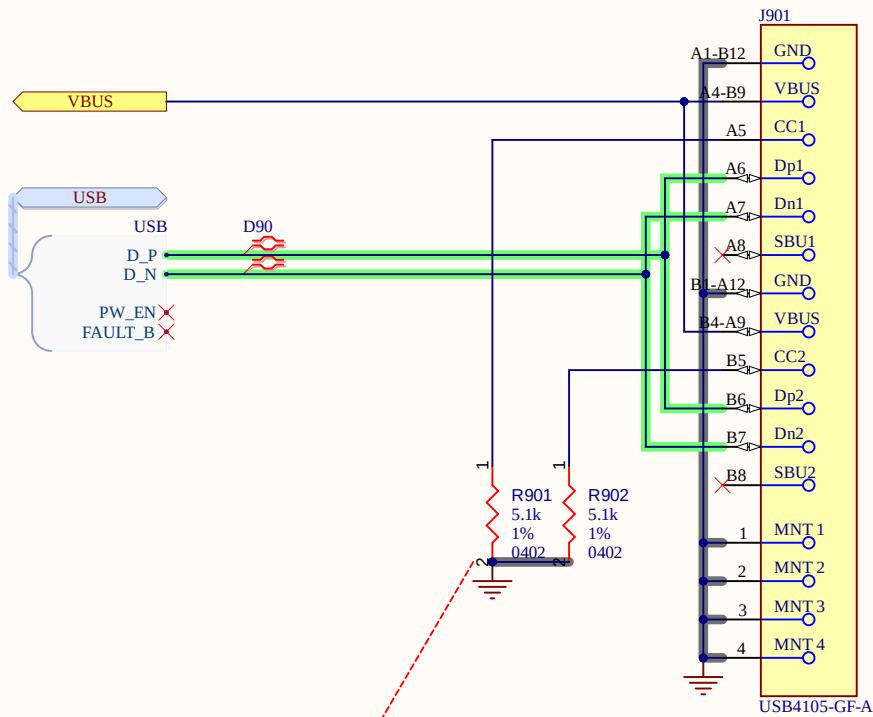
Mode Configuration Straps	
CFG_SEL[1..0]	Description
00	Default Configuration Strap Options Enabled, Self-Powered Operation Enabled, Individual Power Switching & Individual Over-Current Sensing
01	SMBus Slave Configuration Strap Options Disabled, All Registers Configured Over SMBus
10	Default Configuration Strap Options Enabled, Bus-Powered Operation Enabled, Individual Power Switching & Individual Over-Current Sensing
11	I²C EEPROM Configuration Strap Options Disabled, All Registers Configured By I ² C EEPROM

NON_REMx Mode Configuration Straps	
NON_REM[1..0]	Description
00	All Downstream Ports Are Removable
01	Port 1 Is Non-Removable (Permanently Attached)
10	Port 1 & Port 2 Are Non-Removable (Permanently Attached)
11	Port 1, Port 2 & Port 3 Are Non-Removable (Permanently Attached)



A The power distribution switch must have an active high enable input and an over-current fault indication active low output.





A1	GND	B12	GND
A4	VBUS	B9	VBUS
A5	CC1	B8	SBU2
A6	DP1	B7	DN2
A7	DN1	B6	DP2
A8	SBU1	B5	CC2
A9	VBUS	B4	VBUS
A12	GND	B1	GND
PIN	SIGNAL NAME	PIN	SIGNAL NAME

TABLE 7: VALID UFP RD PULL-DOWN RESISTOR VALUES

Rd Implementation	Nominal Value	Detect Power Capability?	Current Source to 1.7V - 5.5V
± 20% voltage clamp	1.1V	No	1.32V
± 20% resistor to GND	5.1kΩ	No	2.18V
± 10% resistor to GND	5.1kΩ	Yes	2.04V

